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EXAMPLE 5.19

A jester in a circus is standing with his arms extended on a turn table rotating with angular velocity ω. He brings his arms closer to his body so that his moment of inertia is reduced to one third of the original value. Find his new angular velocity. [Given: There is no external torque on the turn table in the given situation.]

Solution

Let the moment of inertia of the jester with his arms extended be I. As there is no external torque acting on the jester and the turn table, his total angular momentum is conserved. We can write the equation,

$$I_i\omega_i = I_f\omega_f$$

$$I\omega = \frac{1}{3}I\omega_f \quad \because\left(I_f = \frac{1}{3}I\right)$$

$$\omega_f = 3\omega$$

The above result tells that the final angular velocity is three times that of initial angular velocity.

5.5.3 Work done by Torque

Let us consider a rigid body rotating about a fixed axis. Figure 5.29 shows a point P on the body rotating about an axis perpendicular to the plane of the page. A tangential force F is applied on the body.

It produces a small displacement ds on the body. The work done (dw) by the force is,

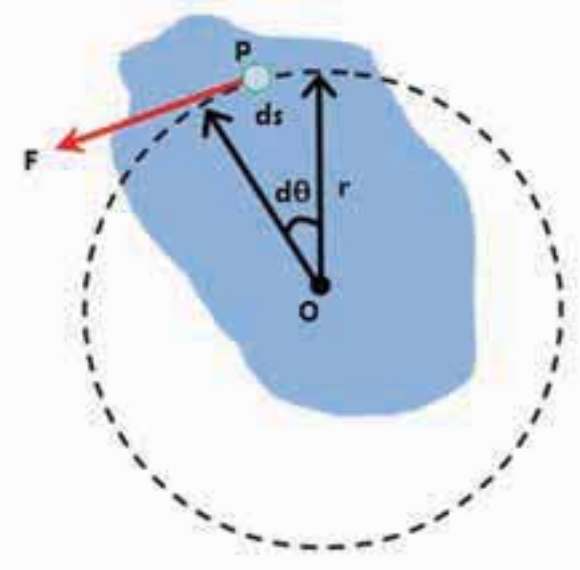


Figure 5.29 Work done by torque

$$dw = F\,ds$$

As the distance ds, the angle of rotation dθ and radius r are related by the expression,

$$ds = r\,d\theta$$

The expression for work done now becomes,

$$dw = F\,ds; \quad dw = F\,r\,d\theta$$

The term (Fr) is the torque τ produced by the force on the body.

$$dw = \tau\,d\theta \qquad (5.51)$$

This expression gives the work done by the external torque τ, which acts on the body rotating about a fixed axis through an angle dθ.

The corresponding expression for work done in translational motion is,

$$dw = F\,ds$$